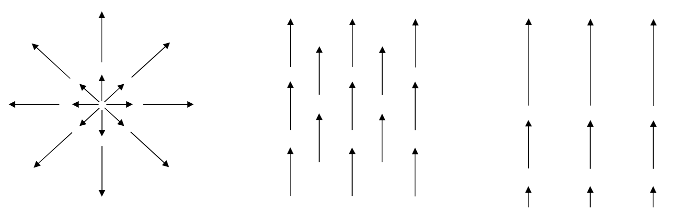
Curl: Example



Home Work: Calculate curl for each field from first principle

1

Important formulae with gradient/divergence/curl

$$\vec{\nabla} \equiv \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}$$

is a vector operator – a hybrid entity satisfying two sets of rule:

(i) vector rule

(ii) partial differentiation rule

$$\vec{A}(\vec{\nabla}\cdot\vec{B}) \neq (\vec{A}\cdot\vec{\nabla})\vec{B}$$

$$\vec{\nabla}(f+g) = \vec{\nabla}f + \vec{\nabla}g$$

$$\vec{\nabla}\cdot(\vec{A}+\vec{B}) = \vec{\nabla}\cdot\vec{A} + \vec{\nabla}\cdot\vec{B}$$

$$\vec{\nabla}\times(\vec{A}+\vec{B}) = \vec{\nabla}\times\vec{A} + \vec{\nabla}\times\vec{B}$$

2

Important formulae with gradient/divergence/curl

$$\vec{\nabla}(fg) = f\,\vec{\nabla}g + g\vec{\nabla}f$$

$$\vec{\nabla}(\vec{A}\cdot\vec{B}) = (\vec{A}\cdot\vec{\nabla})\vec{B} + (\vec{B}\cdot\vec{\nabla})\vec{A} + \vec{A}\times(\vec{\nabla}\times\vec{B}) + \vec{B}\times(\vec{\nabla}\times\vec{A})$$

$$\neq \vec{A}\vec{\nabla}\cdot\vec{B} + \vec{B}\vec{\nabla}\cdot\vec{A}$$

$$\vec{A}\times(\vec{\nabla}\times\vec{B}) = \vec{\nabla}(\vec{A}\cdot\vec{B}) - (\vec{A}\cdot\vec{\nabla})\vec{B}$$

$$\vec{B}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{A}\cdot\vec{B}) - (\vec{B}\cdot\vec{\nabla})\vec{A}$$

$$\vec{\nabla}(\vec{A}\cdot\vec{B}) = (\vec{A}\cdot\vec{\nabla})\vec{B} + \vec{A}\times(\vec{\nabla}\times\vec{B})$$

$$\vec{\nabla}(\vec{A}\cdot\vec{B}) = (\vec{B}\cdot\vec{\nabla})\vec{A} + \vec{B}\times(\vec{\nabla}\times\vec{A})$$

$$\vec{\nabla}\cdot(f\vec{A}) = f\,\vec{\nabla}\cdot\vec{A} + \vec{\nabla}f\cdot\vec{A}$$

$$\vec{\nabla}\cdot(\vec{A}\times\vec{B}) = \vec{B}\cdot(\vec{\nabla}\times\vec{A}) - \vec{A}\cdot(\vec{\nabla}\times\vec{B})$$

$$\vec{\nabla}\times(f\vec{A}) = f\,\vec{\nabla}\times\vec{A} + \vec{\nabla}f\times\vec{A}$$

$\vec{\nabla}\times(\vec{A}\times\vec{B})$ **Home Work**

3

Double derivative of a vector function

Applying ∇ twice, we can construct second derivative

Case 1 $\vec{\nabla}\phi$

Divergence of gradient $\vec{\nabla}\cdot\vec{\nabla}\phi = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)\cdot\left(\hat{x}\frac{\partial\phi}{\partial x} + \hat{y}\frac{\partial\phi}{\partial y} + \hat{z}\frac{\partial\phi}{\partial z}\right)$

$$= \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

$$\boxed{\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}}$$ **Laplacian operator**

Curl of gradient $\boxed{\vec{\nabla}\times\vec{\nabla}\phi \;= 0}$ $\vec{\nabla}\phi$ **is irrotational**

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \partial\phi/\partial x & \partial\phi/\partial y & \partial\phi/\partial z \end{vmatrix} = \hat{x}\left(\frac{\partial}{\partial y}\frac{\partial\varphi}{\partial z} - \frac{\partial}{\partial z}\frac{\partial\varphi}{\partial y}\right) - \hat{y}\left(\frac{\partial}{\partial x}\frac{\partial\varphi}{\partial z} - \frac{\partial}{\partial z}\frac{\partial\varphi}{\partial x}\right) + \ldots$$

$$\frac{\partial^2\phi}{\partial y\partial z} = \frac{\partial^2\phi}{\partial z\partial y}$$

$$\frac{\partial^2\varphi}{\partial x\partial z} = \frac{\partial^2\varphi}{\partial z\partial x}$$

4

Double derivative of a vector function

Applying ∇ twice, we can construct second derivative

Case 2 $\vec{\nabla}\cdot\vec{A}$

Gradient of divergence $\vec{\nabla}(\vec{\nabla}\cdot\vec{A}) \neq (\vec{\nabla}\cdot\vec{\nabla})\vec{A}$

Case 3 $\vec{\nabla}\times\vec{A}$

Divergence of curl $\boxed{\vec{\nabla}\cdot(\vec{\nabla}\times\vec{A}) = 0}$ $\vec{\nabla}\times\vec{A}$ **is solenoidal**

Curl of curl $\boxed{\vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - \nabla^2\vec{A}}$

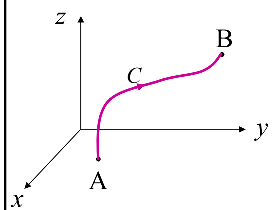
5

Vector Integration

In ED, we encounter several different kinds of integrations

(i) line integral
(ii) surface integral
(iii) volume integral

Line Integral



$$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$$

$$\int_C \phi\, d\vec{l} \qquad \boxed{\int_C \vec{V}\cdot d\vec{l}} \qquad \int_C \vec{V}\times d\vec{l}$$

C : contour $\longrightarrow$ open contour

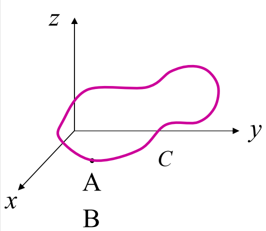
6

Vector Integration

In ED, we encounter several different kinds of integrations

(i) line integral
(ii) surface integral
(iii) volume integral

Line Integral



$$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$$

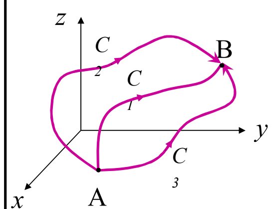
$$\int_C \phi\, d\vec{l} \qquad \boxed{\int_C \vec{V}\cdot d\vec{l}} \qquad \int_C \vec{V}\times d\vec{l}$$

C : contour $\longrightarrow$ open contour
$\longrightarrow$ closed contour $\oint_C \vec{V}\cdot d\vec{l}$

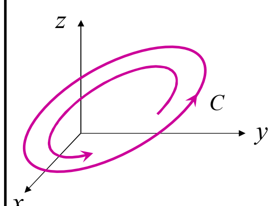
7

Vector Integration

Line Integral



$\int_C \vec{V}\cdot d\vec{l}$ In general path dependent



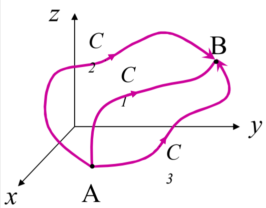
We must define direction of traversal around the loop

Anti-clockwise

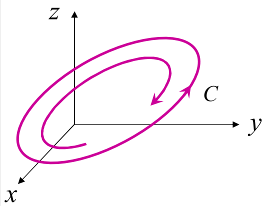
8

Vector Integration

Line Integral



$\int_C \vec{V} \cdot d\vec{l}$ In general path dependent



We must define direction of traversal around the loop

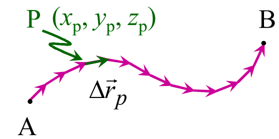
Anti-clockwise

$\int_C \vec{V} \cdot d\vec{l}$ changes its sign

9

Vector Integration

Line Integral



$$\int_C \vec{V} \cdot d\vec{l} = \lim_{N\to\infty} \sum_{p=1}^{N} \vec{V}(x_p, y_p, z_p) \cdot \Delta\vec{r}_p$$

There exists a class of vector field for which line integral is independent of the path followed → conservative vector field

10

Vector Integration

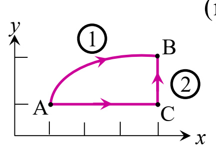
Line Integral: Example

$\int_C \vec{V} \cdot d\vec{l} = ?$ where $\vec{V} = (x+y)\,\hat{x} + (y-x)\,\hat{y}$

along (i) parabola $y^2 = x$ from (1,1) to (4,2)

(ii) $y = 1$ from (1,1) to (4,1) followed by $x = 4$ from (4,1) to (4,2)

(iii) $\oint_C \vec{V} \cdot d\vec{l}$ for the loop ACBA



$d\vec{l} = dx\,\hat{x} + dy\,\hat{y}$

(i) $I = \int_{(1,1)}^{(4,2)} \left((x+y)\,\hat{x} + (y-x)\,\hat{y}\right) \cdot \left(dx\,\hat{x} + dy\,\hat{y}\right)$

$= \int_{(1,1)}^{(4,2)} (x+y)\,dx + (y-x)\,dy$

but $y^2 = x \rightarrow 2y\,dy = dx$ & y: $1 \rightarrow 2$

$I = \int_{y=1}^{y=2} (y^2 + y)\,2y\,dy + (y - y^2)\,dy = \boxed{34/3}$

11

Vector Integration

Line Integral: Example

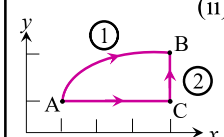
$\int_C \vec{V} \cdot d\vec{l} = ?$ where $\vec{V} = (x+y)\,\hat{x} + (y-x)\,\hat{y}$

along (i) parabola $y^2 = x$ from (1,1) to (4,2)

(ii) $y = 1$ from (1,1) to (4,1) followed by $x = 4$ from (4,1) to (4,2)

(iii) $\oint_C \vec{V} \cdot d\vec{l}$ for the loop ACBA

(ii) Along the horizontal segments, $y = 1, z = 0, x: 1 \rightarrow 4$



$d\vec{l} = dx\,\hat{x} + dy\,\hat{y}$

$\rightarrow dy = dz = 0 \quad d\vec{l} = dx\,\hat{x}$

$\vec{V} \cdot d\vec{l} = (x+y)dx = (x+1)dx$

$I_1 = \int_{x=1}^{x=4} (x+1)\,dx = \boxed{21/2}$

12

Vector Integration

Line Integral: Example

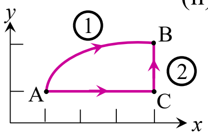
$$\boxed{I_1 = 21/2}$$

$\int_C \vec{V}\cdot d\vec{l} = ?$ where $\vec{V} = (x+y)\,\hat{x} + (y-x)\,\hat{y}$

along (i) parabola $y^2 = x$ from (1,1) to (4,2)

(ii) $y = 1$ from (1,1) to (4,1) followed by $x = 4$ from (4,1) to (4,2)

(iii) $\oint_C \vec{V}\cdot d\vec{l}$ for the loop ACBA



$d\vec{l} = dx\,\hat{x} + dy\,\hat{y}$

(ii) Along the vertical segments, $x = 4, z = 0, y: 1 \to 2$

$\to dx = dz = 0 \quad d\vec{l} = dy\,\hat{y}$

$\vec{V}\cdot d\vec{l} = (y-x)dy \quad = (y-4)dy$

$I_2 = \int_{y=1}^{y=2} (y-4)\,dy = \boxed{-5/2}$

$$\boxed{I_1 + I_2 = 8}$$

13

Vector Integration

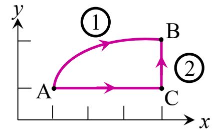
Line Integral: Example

$\int_C \vec{V}\cdot d\vec{l} = ?$ where $\vec{V} = (x+y)\,\hat{x} + (y-x)\,\hat{y}$

along (i) parabola $y^2 = x$ from (1,1) to (4,2)

(ii) $y = 1$ from (1,1) to (4,1) followed by $x = 4$ from (4,1) to (4,2)

(iii) $\oint_C \vec{V}\cdot d\vec{l}$ for the loop ACBA



$d\vec{l} = dx\,\hat{x} + dy\,\hat{y}$

(iii) Along path "1", $\int_C \vec{V}\cdot d\vec{l} = 34/3$

Along path "2", $\int_C \vec{V}\cdot d\vec{l} = 8$

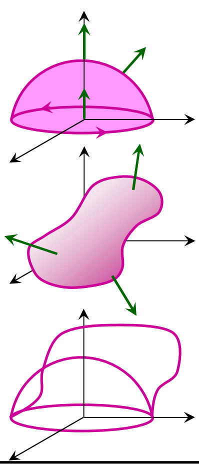
$$\to \boxed{\oint_C \vec{V}\cdot d\vec{l} = 8 - 34/3 = -10/3 \neq 0}$$

14

Vector Integration

Surface Integral:

$\int_S \phi\, d\vec{a}$ $\quad \boxed{\int_S \vec{V}\cdot d\vec{a}}$ $\quad \int_S \vec{V}\times d\vec{a}$



S : surface element $\to$ open element

$\to$ closed element $\oint_S \vec{V}\cdot d\vec{a}$

$d\vec{a} = da\,\hat{n}$ $\quad da$: area of the element

$\hat{n}$: unit vector

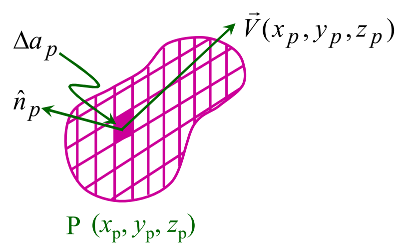
$\int_S \vec{V}\cdot d\vec{a}$ in general, depends on the choice of surface

There exists a class of vector field for which surface integral is independent of the choice of the surface

15

Vector Integration

Surface Integral:

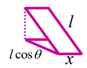


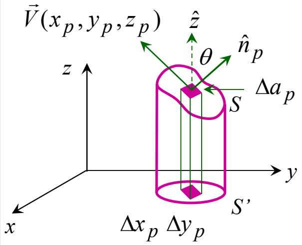
$$\int_S \vec{V}\cdot d\vec{a} = \lim_{N\to\infty} \sum_{p=1}^{N} \vec{V}(x_p, y_p, z_p)\cdot \Delta a_p\, \hat{n}_p$$

16

Vector Integration

Surface Integral: Evaluation formula





$$\Delta x_p\ \Delta y_p = \Delta a_p \cos\theta$$

$$= \Delta a_p \left|\hat{n}_p \cdot \hat{z}\right|$$

$$\longrightarrow \quad \Delta a_p = \frac{\Delta x_p\ \Delta y_p}{\left|\hat{n}_p \cdot \hat{z}\right|}$$

$$\int_S \vec{V}\cdot d\vec{a} = \lim_{N\to\infty}\sum_{p=1}^{N} \vec{V}_p \cdot \hat{n}_p \Delta a_p$$

$$= \lim_{N\to\infty}\sum_{p=1}^{N} \vec{V}_p \cdot \hat{n}_p \frac{\Delta x_p\ \Delta y_p}{\left|\hat{n}_p \cdot \hat{z}\right|}$$

$\hat{n} = \textbf{grad}\,\phi / \left|\textbf{grad}\phi\right|$

$S:\ \phi(x,y,z) = c$

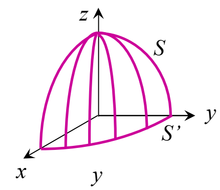
$$\boxed{\int_S \vec{V}\cdot d\vec{a} = \int_x\int_y \vec{V}\cdot\hat{n}\frac{dxdy}{\left|\hat{n}\cdot\hat{z}\right|}}$$

17

Surface Integration: Example 1

$\int_s \vec{V}\cdot\hat{n}\, da = ?\quad \vec{V} = (yz\hat{x} + zx\hat{y} + xy\hat{z})\qquad S:\ x^2+y^2+z^2=1$

$\phi(x,y,z) = x^2+y^2+z^2 = \text{const}$

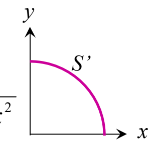


$$\int_s \vec{V}\cdot\hat{n}\, da = \int_x\int_y \vec{V}\cdot\hat{n}\frac{dxdy}{\left|\hat{n}\cdot\hat{z}\right|}$$

$$\hat{n} = \frac{\nabla\phi}{\left|\nabla\phi\right|} = \frac{2x\hat{x}+2y\hat{y}+2z\hat{z}}{\sqrt{4x^2+4y^2+4z^2}} = x\hat{x}+y\hat{y}+z\hat{z}$$

$\longrightarrow \hat{n}\cdot\hat{z} = z \quad \vec{V}\cdot\hat{n} = 3xyz$

$x: 0 \to 1,$
$y: 0 \to \sqrt{1-x^2}$



$$\int_s \vec{V}\cdot\hat{n}\, da = \int_x\int_y 3xyz\frac{dxdy}{z} = \int_x\int_y 3xy\, dxdy$$

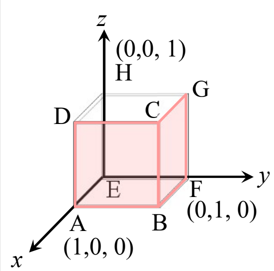
$$= \int_{x=0}^{x=1}\int_{y=0}^{y=\sqrt{1-x^2}} 3xy\, dxdy = 3\int_{x=0}^{x=1} x\left(\int_{y=0}^{y=\sqrt{1-x^2}} y\, dy\right)dx = \frac{3}{2}\int_{x=0}^{x=1} x\left(1-x^2\right)dx$$

$= 3/8$

18

Surface Integration: Example 2

$\int_s \vec{V}\cdot\hat{n}\, da = ?\quad \vec{V} = (yz\hat{x} + zx\hat{y} + xy\hat{z})\qquad S:$ Unit cube



For ABCDA: $d\vec{a} = dydz\,\hat{x}$

$\vec{V}\cdot d\vec{a}\big|_{ABCDA} = yz\, dydz\quad (y: 0\to 1, z: 0\to 1)$

$$\int_{ABCDA} \vec{V}\cdot\hat{n}\, da = \int_{y=0}^{1}\int_{z=0}^{1} yzdydz = \int_{y=0}^{1} ydy\int_{z=0}^{1} zdz$$

$$= \left(\frac{y^2}{2}\right)_0^1\left(\frac{z^2}{2}\right)_0^1 = 1/4$$

For BFGCB: $d\vec{a} = dxdz\,\hat{y}$

$\vec{V}\cdot d\vec{a}\big|_{BFGCB} = zx\, dxdz\quad (x: 0\to 1, z: 0\to 1)$

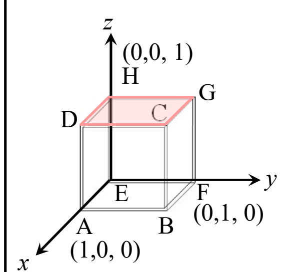
$$\int_{BFGCB} \vec{V}\cdot\hat{n}\, da = \int_{z=0}^{1}\int_{x=0}^{1} zxdzdx = \int_{z=0}^{1} zdz\int_{x=0}^{1} xdx$$

$= 1/4$

19

Surface Integration: Example 2

$\int_s \vec{V}\cdot\hat{n}\, da = ?\quad \vec{V} = (yz\hat{x} + zx\hat{y} + xy\hat{z})\qquad S:$ Unit cube



For CGHDC: $d\vec{a} = dxdy\,\hat{z}$

$\vec{V}\cdot d\vec{a}\big|_{CGHDC} = xy\, dxdy\quad (x: 0\to 1, y: 0\to 1)$

$$\int_{CGHDC} \vec{V}\cdot\hat{n}\, da = \int_{x=0}^{1}\int_{y=0}^{1} xydxdy = \int_{x=0}^{1} xdx\int_{y=0}^{1} ydy$$

$= 1/4$ • • •

20

Volume Integration

$$\int_V \phi \, dV$$

$$\int_V \vec{F} \, dV$$

$$dV = dx\, dy\, dz$$

Essentially a triple integral

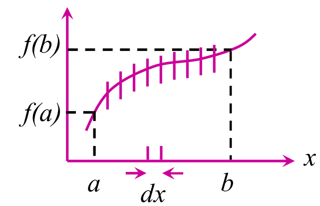
21

Fundamental Theorem of Calculus

$$\frac{df}{dx} = F(x)$$

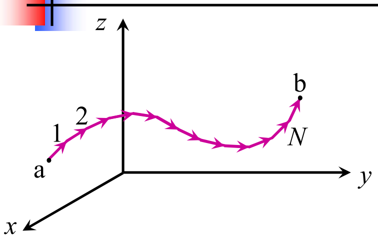
FTC $$\int_a^b \frac{df}{dx} dx = f(b) - f(a)$$

$\frac{df}{dx} dx = df$ infinitesimal change in f with infinitesimal change in x



22

Fundamental Theorem … $\int_a^b \frac{df}{dx} dx = f(b) - f(a)$



$$\phi(x, y, z)$$

$$d\phi = \vec{\nabla}\phi \cdot d\vec{l}$$

$$d\phi_1 = \vec{\nabla}\phi_1 \cdot d\vec{l}_1 = \phi(1) - \phi(a)$$

$$d\phi_2 = \vec{\nabla}\phi_2 \cdot d\vec{l}_2 = \phi(2) - \phi(1)$$

$$d\phi_N = \vec{\nabla}\phi_N \cdot d\vec{l}_N = \phi(b) - \phi(N)$$

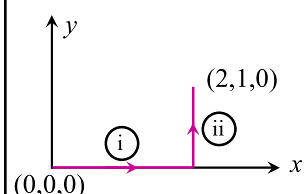
$$d\phi_1 + d\phi_2 + \ldots + d\phi_N = \vec{\nabla}\phi_1 \cdot d\vec{l}_1 + \vec{\nabla}\phi_2 \cdot d\vec{l}_2 + \ldots + \vec{\nabla}\phi_N \cdot d\vec{l}_N = \phi(b) - \phi(a)$$

with $N \to \infty$ $$\boxed{\int_a^b \vec{\nabla}\phi \cdot d\vec{l} = \phi(b) - \phi(a)} \longrightarrow \oint_L \vec{\nabla}\phi \cdot d\vec{l} = 0$$

Fundamental Theorem of Gradient

23

Fundamental Theorem of Gradient: Example



Consider a scalar function $\phi = xy^2$ and points a (0,0,0) & b (2,1,0)

Using FTOG

$$\int_a^b \vec{\nabla}\phi \cdot d\vec{l} = \phi(b) - \phi(a) = 2 \cdot 1^2 - 0 = 2$$

Path 1: For (i), $x : 0-2,\ y = 0,\ dy = dz = 0$ $\quad d\vec{l} = dx\,\hat{x}$

$$\vec{\nabla}\phi \cdot d\vec{l} = \left(y^2\,\hat{x} + 2yx\hat{y}\right) \cdot dx\,\hat{x} = y^2 dx \longrightarrow \int_i \vec{\nabla}\phi \cdot d\vec{l} = 0$$

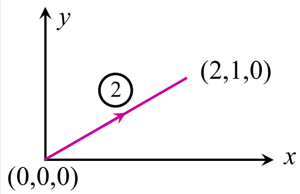
Path 1: For (ii), $x : 2,\ y : 0-1,\ dx = dz = 0$ $\quad d\vec{l} = dy\,\hat{y}$

$$\vec{\nabla}\phi \cdot d\vec{l} = \left(y^2\,\hat{x} + 2yx\hat{y}\right) \cdot dy\,\hat{y} = 2yx\,dy \longrightarrow \int_{ii} \vec{\nabla}\phi \cdot d\vec{l} = \int_0^1 4y\,dy = 2$$

$$\longrightarrow \int_1 \vec{\nabla}\phi \cdot d\vec{l} = \int_i \vec{\nabla}\phi \cdot d\vec{l} + \int_{ii} \vec{\nabla}\phi \cdot d\vec{l} = 2$$

24

Fundamental Theorem of Gradient: Example



Consider a scalar function $\phi = xy^2$ and points a (0,0,0) & b (2,1,0)

Using FTOG

$$\int_a^b \vec{\nabla}\phi \cdot d\vec{l} = \phi(b) - \phi(a) = 2\cdot 1^2 - 0 = 2$$

Path 2: $y = \mathrm{m}\,x + \mathrm{c}$, $\mathrm{m} = 1/2$, $\mathrm{c} = 0$, $y = x/2$, $dy = dx/2$

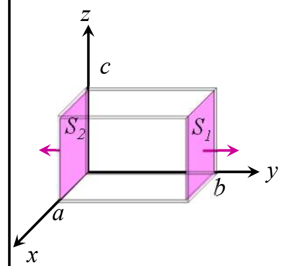
$$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} \qquad \vec{\nabla}\phi = \left(y^2\,\hat{x} + 2yx\hat{y}\right)$$

$$\vec{\nabla}\phi\cdot d\vec{l} = y^2\,dx + 2yx\,dy = \frac{x^2}{4}dx + 2\frac{x}{2}x\frac{dx}{2} = \frac{3}{4}x^2 dx$$

$$\int_2 \vec{\nabla}\phi\cdot d\vec{l} = \int_0^2 \frac{3}{4}x^2 dx = 2$$

Fundamental Theorem of Divergence/Green/Gauss

$$\int_{volume}\left(\vec{\nabla}\cdot\vec{V}\right)d\tau = \oint_{surface}\vec{V}\cdot d\vec{a}$$



$$\int_{volume}\left(\vec{\nabla}\cdot\vec{V}\right)d\tau = \int_0^c\int_0^b\int_0^a\left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}\right)dx\,dy\,dz$$

$$\int_0^c\int_0^a\left(\int_0^b \frac{\partial V_y}{\partial y}dy\right)_{const\ x,z} dx\,dz$$

$$= \int_0^c\int_0^a [V_y(x, y=b, z) - V_y(x, y=0, z)]\,dx\,dz$$

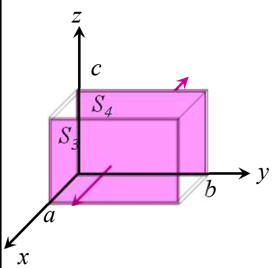
$$= \int_0^c\int_0^a V_y(x, y=b, z)\,dx\,dz - \int_0^c\int_0^a V_y(x, y=0, z)\,dx\,dz$$

$$= \int_0^c\int_0^a \vec{V}(y=b)\cdot\hat{y}\,dx\,dz + \int_0^c\int_0^a \vec{V}(y=0)\cdot(-\hat{y}\,dx\,dz)$$

$$= \int_{S_1}\vec{V}\cdot d\vec{a} + \int_{S_2}\vec{V}\cdot d\vec{a}$$

Fundamental Theorem of Divergence/Green/Gauss

$$\int_{volume}\left(\vec{\nabla}\cdot\vec{V}\right)d\tau = \oint_{surface}\vec{V}\cdot d\vec{a}$$



$$\int_{volume}\left(\vec{\nabla}\cdot\vec{V}\right)d\tau = \int_0^c\int_0^b\int_0^a\left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}\right)dx\,dy\,dz$$

$$\int_0^b\int_0^c\left(\int_0^a \frac{\partial V_x}{\partial x}dx\right)_{const\ y,z} dy\,dz$$

$$= \int_0^b\int_0^c [V_x(x=a, y, z) - V_x(x=0, y, z)]\,dy\,dz$$

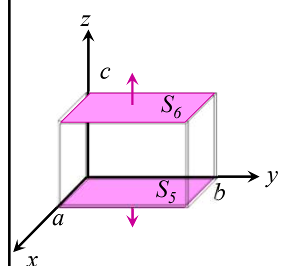
$$= \int_0^b\int_0^c V_x(x=a, y, z)\,dx\,dz - \int_0^b\int_0^c V_x(x=0, y, z)\,dx\,dz$$

$$= \int_0^b\int_0^c \vec{V}(x=a)\cdot\hat{x}\,dy\,dz + \int_0^b\int_0^c \vec{V}(x=0)\cdot(-\hat{x}\,dy\,dz)$$

$$= \int_{S_3}\vec{V}\cdot d\vec{a} + \int_{S_4}\vec{V}\cdot d\vec{a}$$

Fundamental Theorem of Divergence/Green/Gauss

$$\int_{volume}\left(\vec{\nabla}\cdot\vec{V}\right)d\tau = \oint_{surface}\vec{V}\cdot d\vec{a}$$



$$\int_{volume}\left(\vec{\nabla}\cdot\vec{V}\right)d\tau = \int_0^c\int_0^b\int_0^a\left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}\right)dx\,dy\,dz$$

$$\int_0^a\int_0^b\left(\int_0^c \frac{\partial V_z}{\partial z}dz\right)_{const\ x,y} dx\,dy$$

$$= \int_{S_5}\vec{V}\cdot d\vec{a} + \int_{S_6}\vec{V}\cdot d\vec{a}$$

Fundamental Theorem of Divergence/Green/Gauss

$$\boxed{\int_{volume}\left(\vec{\nabla}\cdot\vec{V}\right)d\tau = \oint_{surface}\vec{V}\cdot d\vec{a}}$$

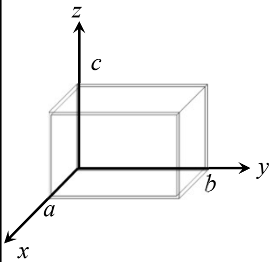
$$\int_{volume}\left(\vec{\nabla}\cdot\vec{V}\right)d\tau = \int_0^c\int_0^b\int_0^a\left(\frac{\partial V_x}{\partial x}+\frac{\partial V_y}{\partial y}+\frac{\partial V_z}{\partial z}\right)dx\,dy\,dz$$

$$= \int_{S_1}\vec{V}\cdot d\vec{a}+\int_{S_2}\vec{V}\cdot d\vec{a}$$

$$+ \int_{S_5}\vec{V}\cdot d\vec{a}+\int_{S_6}\vec{V}\cdot d\vec{a}$$

$$+ \int_{S_3}\vec{V}\cdot d\vec{a}+\int_{S_4}\vec{V}\cdot d\vec{a}$$

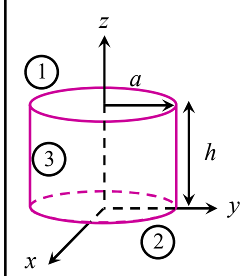
$$= \oint_S \vec{V}\cdot d\vec{a}$$



29

Fundamental Theorem of Divergence: Example

Consider a vector function $\vec{V}=\vec{r}$ Evaluate $\oint \vec{V}\cdot d\vec{a}$



$$\oint_S \vec{V}\cdot d\vec{a} = \int_{S_1}\vec{V}\cdot d\vec{a}+\int_{S_2}\vec{V}\cdot d\vec{a}+\int_{S_3}\vec{V}\cdot d\vec{a} = \boxed{3\pi a^2 h}$$

For S_1: $\hat{n}=\hat{z}$, $z=h$ $d\vec{a}=\hat{z}\,da$

$\vec{V}\cdot d\vec{a} = z\,da = h\,da \rightarrow \int_{S_1}\vec{V}\cdot d\vec{a} = h\int da = \underline{h\pi a^2}$

For S_2: $\hat{n}=-\hat{z}$, $z=0$ $d\vec{a}=-\hat{z}\,da$ $\vec{V}\cdot d\vec{a}=-z\,da$ $=0$

$\rightarrow \int_{S_2}\vec{V}\cdot d\vec{a} = \underline{0}$

For S_3: $\vec{n}=?$ $\vec{n}=x\hat{x}+y\hat{y}$ $\hat{n}=\dfrac{x\hat{x}+y\hat{y}}{\sqrt{x^2+y^2}} = \dfrac{1}{a}(x\hat{x}+y\hat{y})$

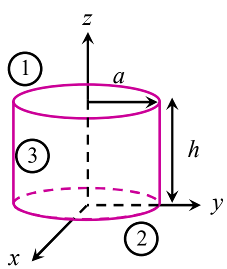
$\int_{S_3}\vec{V}\cdot d\vec{a} = \dfrac{1}{a}\int(x^2+y^2)\,da = \dfrac{a^2}{a}$ (area of the cylindrical surface)

$= a\cdot 2\pi ah = \underline{2\pi a^2 h}$

30

Fundamental Theorem of Divergence: Example

Consider a vector function $\vec{V}=\vec{r}$ Evaluate $\oint \vec{V}\cdot d\vec{a}$



$$\oint_S \vec{V}\cdot d\vec{a} = \int_{S_1}\vec{V}\cdot d\vec{a}+\int_{S_2}\vec{V}\cdot d\vec{a}+\int_{S_3}\vec{V}\cdot d\vec{a} = \boxed{3\pi a^2 h}$$

Using FTOD:

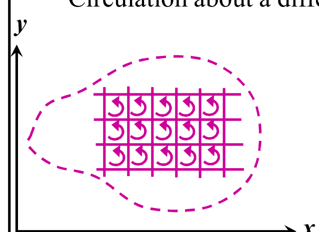
$$\oint_S \vec{V}\cdot d\vec{a} = \int_V(\vec{\nabla}\cdot\vec{V})\,d\tau = 3\int_V d\tau = 3\pi a^2 h$$

31

Fundamental Theorem of Curl/Stokes Theorem

$$\boxed{\int_S\left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a} = \oint_L \vec{V}\cdot d\vec{l}}$$

Circulation about a differential rectangle: $\left(\vec{\nabla}\times\vec{V}\right)_z dxdy$

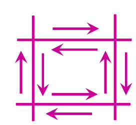


or $\displaystyle\sum_{four\ sides}\vec{V}\cdot d\vec{l} = \left(\vec{\nabla}\times\vec{V}\right)_z dxdy$

$= \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a}$ $\because d\vec{a}=\hat{z}\,dxdy$

$$\sum_N\sum_{four\ sides}\vec{V}\cdot d\vec{l} = \sum_N\left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a}$$

$$\sum_{exterior\ line\ segment}\vec{V}\cdot d\vec{l} = \sum_{rect}\left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a}$$



$$\oint_L \vec{V}\cdot d\vec{l} = \int_S\left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a}$$ **FTOC**

32

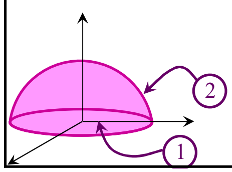
Fundamental Theorem of Curl/Stokes Theorem

$$\boxed{\int_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a} = \oint_L \vec{V}\cdot d\vec{l}}$$

Surface integral of a vector function, in general, depends critically on the surface we choose to integrate.

That's not true for the curls.

$\int_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a}$ is equal to the line integral of $\vec{V}$ around the boundary which does not make any reference of the specific surface we choose



$\int_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a}$ depends only on the boundary

$\longrightarrow$ independent of surface chosen

$\longrightarrow \oint_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a} = 0$

33

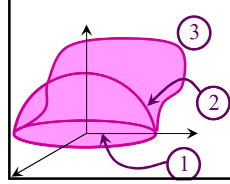
Fundamental Theorem of Curl/Stokes Theorem

$$\boxed{\int_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a} = \oint_L \vec{V}\cdot d\vec{l}}$$

Surface integral of a vector function, in general, depends critically on the surface we choose to integrate.

That's not true for the curls.

$\int_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a}$ is equal to the line integral of $\vec{V}$ around the boundary which does not make any reference of the specific surface we choose



$\int_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a}$ depends only on the boundary

$\longrightarrow$ independent of surface chosen

$\longrightarrow \oint_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a} = 0$

34

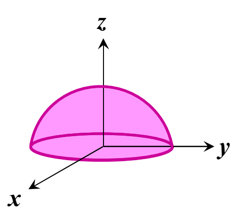
Fundamental Theorem of Curl: Example

Evaluate $\int_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a}$ over a hemisphere $x^2+y^2+z^2=a^2;\ z\ge 0$

for $\vec{V} = 4y\,\hat{x} + x\,\hat{y} + 2z\,\hat{z}$

Boundary of the surface is $x^2+y^2=a^2$

One can choose any surface bounded by $x^2+y^2=a^2$



$\hat{n} = \hat{z}$

$\vec{\nabla}\times\vec{V} = -3\hat{z}$

$\left(\vec{\nabla}\times\vec{V}\right)\cdot\hat{n} = -3$

$$\int_S \left(\vec{\nabla}\times\vec{V}\right)\cdot d\vec{a} = -3\int_S da = -3\pi a^2$$

35

Important Tool: Integration by Part

$$\frac{d}{dx}(fg) = f\frac{dg}{dx} + g\frac{df}{dx}$$

$$\int_a^b \frac{d}{dx}(fg)\,dx = \int_a^b f\left(\frac{dg}{dx}\right)dx + \int_a^b g\left(\frac{df}{dx}\right)dx = (fg)\Big|_a^b$$

$$\boxed{\int_a^b f\left(\frac{dg}{dx}\right)dx = -\int_a^b g\left(\frac{df}{dx}\right)dx + (fg)\Big|_a^b}$$

Example:

$\int_0^\infty xe^{-x}dx = ?\qquad e^{-x} = \frac{d}{dx}(-e^{-x})\qquad f(x) = x\qquad g(x) = -e^{-x}\qquad \frac{df}{dx} = 1$

$$\int_0^\infty xe^{-x}dx = \int_0^\infty x\frac{d}{dx}(e^{-x})dx = -\int_0^\infty (-e^{-x})\frac{dx}{dx}dx + (-xe^{-x})_0^\infty = \int_0^\infty e^{-x}dx - xe^{-x}\Big|_0^\infty$$

$$= -e^{-x}\Big|_0^\infty - (0-0) = 1$$

36

Important Tool: Integration by Part

$$\frac{d}{dx}(fg) = f\frac{dg}{dx} + g\frac{df}{dx}$$

$$\int_a^b f\left(\frac{dg}{dx}\right)dx = -\int_a^b g\left(\frac{df}{dx}\right)dx + (fg)\Big|_a^b$$

$$\vec{\nabla}\cdot\left(f\vec{A}\right) = f\left(\vec{\nabla}\cdot\vec{A}\right) + \left(\vec{\nabla}f\right)\cdot\vec{A}$$

$$\int_{vol}\vec{\nabla}\cdot\left(f\vec{A}\right)d\tau = \int_{vol} f\left(\vec{\nabla}\cdot\vec{A}\right)d\tau + \int_{vol}\left(\vec{\nabla}f\right)\cdot\vec{A}\,d\tau = \oint_S f\vec{A}\cdot d\vec{a}$$

$$\boxed{\int_{vol} f\left(\vec{\nabla}\cdot\vec{A}\right)d\tau = -\int_{vol}\left(\vec{\nabla}f\right)\cdot\vec{A}\,d\tau + \oint_S f\vec{A}\cdot d\vec{a}}$$

37